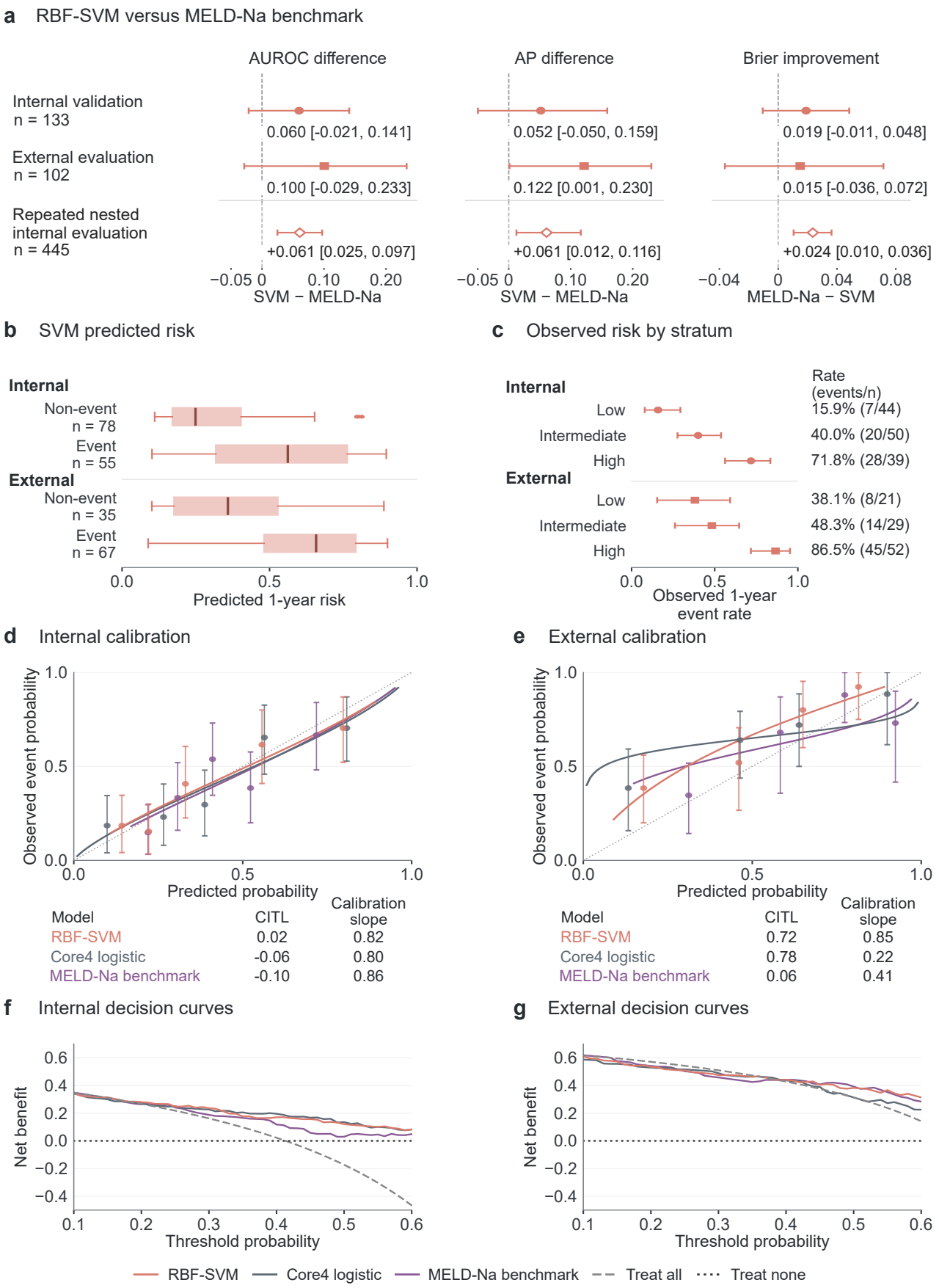

a RBF-SVM versus MELD-Na benchmark
AUROC difference
AP difference
Brier improvement
Internal validation
n = 133
External evaluation
n = 102
Repeated nested
internal evaluation
n = 445
0.060 [-0.021, 0.141]
0.100 [-0.029, 0.233]
+0.061 [0.025, 0.097]
0.052 [-0.050, 0.159]
0.122 [0.001, 0.230]
+0.061 [0.012, 0.116]
0.019 [-0.011, 0.048]
0.015 [-0.036, 0.072]
+0.024 [0.010, 0.036]
−0.05 0 0.10 0.20
SVM − MELD-Na
−0.05 0 0.10 0.20
SVM − MELD-Na
−0.04 0 0.04 0.08
MELD-Na − SVM
b SVM predicted risk
Internal
Non-event
n = 78
Event
n = 55
External
Non-event
n = 35
Event
n = 67
0.0 0.5 1.0
Predicted 1-year risk
c Observed risk by stratum
Rate
(events/n)
Internal
Low
15.9% (7/44)
Intermediate
40.0% (20/50)
High
71.8% (28/39)
External
Low
38.1% (8/21)
Intermediate
48.3% (14/29)
High
86.5% (45/52)
0.0 0.5 1.0
Observed 1-year
event rate
d Internal calibration
Observed event probability
0.0 0.5 1.0
Predicted probability
Model
CITL
Calibration slope
RBF-SVM
0.02
0.82
Core4 logistic
-0.06
0.80
MELD-Na benchmark
-0.10
0.86
e External calibration
Observed event probability
0.0 0.5 1.0
Predicted probability
Model
CITL
Calibration slope
RBF-SVM
0.72
0.85
Core4 logistic
0.78
0.22
MELD-Na benchmark
0.06
0.41
f Internal decision curves
Net benefit
0.6 0.4 0.2 0.0 −0.2 −0.4
0.1 0.2 0.3 0.4 0.5 0.6
Threshold probability
g External decision curves
Net benefit
0.6 0.4 0.2 0.0 −0.2 −0.4
0.1 0.2 0.3 0.4 0.5 0.6
Threshold probability
RBF-SVM
Core4 logistic
MELD-Na benchmark
Treat all
Treat none